TOPSTECHNOLOGIES
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N+ - Network Fundamentals and building network

Wireless Networking

[Q-1] List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Standard	**Wi-Fi Generation**	**Key Difference**
802.11n	Wi-Fi 4	Up to about 600 Mbps, supports both 2.4 GHz and 5 GHz bands
802.11ac	Wi-Fi 5	Faster speeds (up to several Gbps), mainly uses the 5 GHz band
802.11ax	Wi-Fi 6	Higher speed, better efficiency, improved performance in crowded networks, supports 2.4 GHz and 5 GHz

[Q-2] Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

```
Smartphone
   |
 (Wi-Fi)
   |
   v
| Wi-Fi Router|
    |
  LAN Cable
    |
    v
| ISP Modem   |
    |
  Internet
```

Connection Types

- **Wireless:** Smartphone → Wi-Fi Router
- **Wired:** Router → ISP Modem → Internet

[Q-3] Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

SSID	Signal Strength	Security
Home_WIFI	Strong	WPA2
JioFiber_5G	Medium	WPA2
Neighbor_Net	Weak	WPA3

Best Choice: Home_WIFI

Reason:

- Strong signal strength
- Secure encryption (WPA2 or WPA3)
- Better performance and reliability

Avoid connecting to open or unsecured networks whenever possible.

[Q-4] Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID hiding means the Wi-Fi network name is not displayed in the list of available wireless networks. Users must manually enter the network name (SSID) to connect.

Advantages

- Provides a small amount of privacy.
- Casual users cannot easily see the network name.

Disadvantages

- Does not provide strong security.
- Wi-Fi scanning tools can still detect the network.

- Users must manually enter the SSID to connect.
- Can make network management less convenient.